

Uber

Ride System, Database Design

Overview

This project presents a simplified relational database design for a ride system inspired by Uber. A passenger requests a trip, which is fulfilled by a driver using their registered car. The system models the core interactions of such a service through four related classes and the foreign key relationships that connect them.

Driver • Car • Passenger • Trip

Each driver owns one car, and each trip links a passenger, a driver, and the car used for that trip. Constraints such as CHECK, UNIQUE, and NOT NULL are used throughout to keep the data consistent.

Database Schema and SQL code

```
create database uber;

CREATE TABLE driver(
  id INT PRIMARY KEY,
  full_name VARCHAR(30) NOT NULL,
  phone VARCHAR(10) UNIQUE NOT NULL,
  license_number VARCHAR(20) UNIQUE NOT NULL,
  rating DOUBLE DEFAULT 0.0 CHECK ( rating <= 5 ),
  status VARCHAR(10) CHECK ( status = 'available' OR status = 'busy' ),
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE car(
  id INT PRIMARY KEY,
  plate_number VARCHAR(10) UNIQUE NOT NULL,
  model VARCHAR(20) NOT NULL,
  color VARCHAR(15),
  driver_id INT,
  FOREIGN KEY (driver_id) REFERENCES driver(id)
);

CREATE TABLE passenger(
  id INT PRIMARY KEY,
  full_name VARCHAR(30) NOT NULL,
  phone VARCHAR(15) UNIQUE NOT NULL,
  email VARCHAR(30) UNIQUE,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE trip(
  id INT PRIMARY KEY,
  rider_id INT,
  FOREIGN KEY (rider_id) REFERENCES passenger(id),
  driver_id INT,
  FOREIGN KEY (driver_id) REFERENCES driver(id),
  car_id INT,
  FOREIGN KEY (car_id) REFERENCES car(id),
```

```

        pickup_location VARCHAR(50) NOT NULL,
        destination_location VARCHAR(50) NOT NULL,
        status VARCHAR(10) CHECK ( status = 'requested' OR status = 'ongoing'
OR status = 'completed' OR status = 'cancelled' ),
        price INT ,
        arrival_time DATETIME
    );

INSERT INTO driver (id, full_name, phone, license_number, rating, status)
VALUES (1, 'fahad', '0501111111', 'bbb3524', 4.8, 'available');
INSERT INTO driver (id, full_name, phone, license_number, rating, status)
VALUES (2, 'Nasser', '0502222222', 'aaa9562', 4.5, 'available');

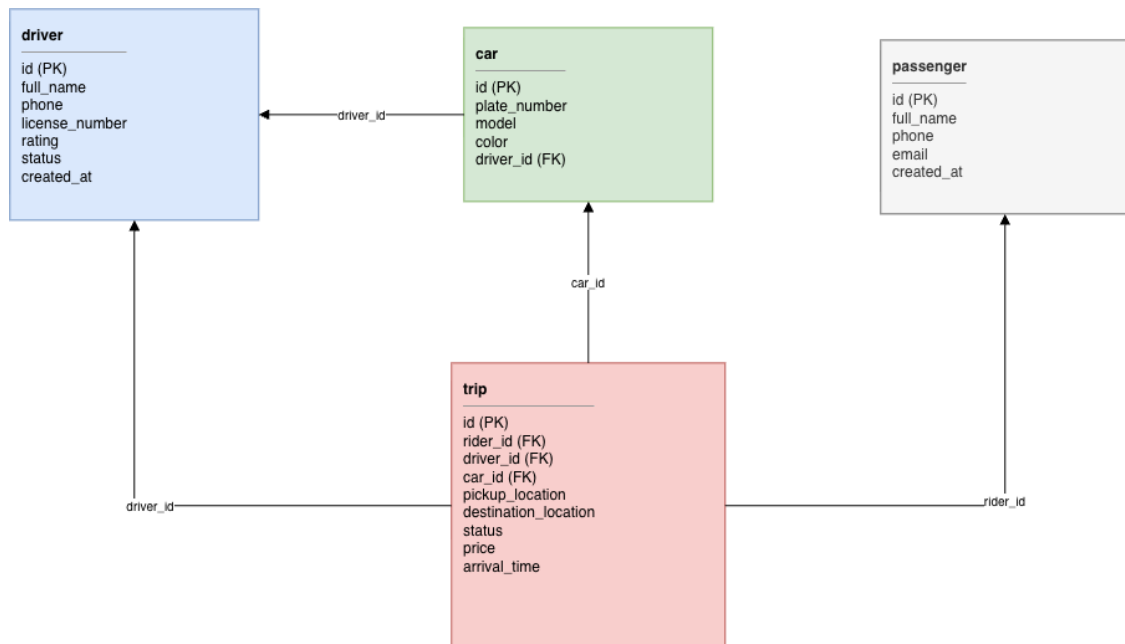
INSERT INTO car (id, plate_number, model, color, driver_id) VALUES (1,
'ABC-123', 'Toyota Camry', 'White', 1);
INSERT INTO car (id, plate_number, model, color, driver_id) VALUES (2,
'XYZ-789', 'Ford territory', 'Gold', 2);

INSERT INTO passenger (id, full_name, phone, email) VALUES (1, 'Fai
Alharthi', '0503333333', 'fai@example.com');
INSERT INTO passenger (id, full_name, phone, email) VALUES (2, 'Noura
Saleh', '0504444444', 'noura@example.com');

INSERT INTO trip (id, rider_id, driver_id, car_id, pickup_location,
destination_location, status, price, arrival_time) VALUES (1, 1, 1, 1,
'King Fahd Road', 'Airport', 'ongoing', 45, '2026-09-19 14:30:00');
INSERT INTO trip (id, rider_id, driver_id, car_id, pickup_location,
destination_location, status, price, arrival_time) VALUES (2, 2, 2, 2,
'Olaya Street', 'Mall', 'requested', 30, '2026-09-19 15:00:00');

UPDATE trip SET status = 'cancelled' WHERE id = 2;
UPDATE driver SET status = 'busy' WHERE id = 1;

```



UML relationships between Driver, Car, Passenger, and Trip.